

Tanim Ahmmmod

B.Sc. in Electrical and Electronic Engineering
Rajshahi University of Engineering & Technology

+880 1954999739

Portfolio | LinkedIn | Scholar

tanim.eee.ruet@gmail.com

Talaimari, Rajshahi 6204, Bangladesh

SUMMARY

I recently graduated in Electrical and Electronic Engineering. Being passionate about research and problem-solving, I am seeking a Master's or PhD program and a Graduate Assistantship to deepen my involvement in research and contribute to academic excellence. My goal is to make a significant impact in application field of medical wearable devices and AI, leveraging my research experience, teamwork, and problem-solving skills for innovative projects.

PROFESSIONAL EXPERIENCE

•SPML Research Lab, RUET

Research Assistant

Sep, 2025 – Present

– Created Dataset of ECG, PPG, and GSR signals from different subjects using Shimmer devices.

•ResearchBuddy AI Lab, CUET

Research Trainee

Jan, 2024 – June, 2024

– Explored LLM and self-supervised learning (SSL) models.

•Udvash Academic & Admission Care

Academic Evaluator & Tutor

Mar, 2022 – Dec, 2023

– Evaluated exam scripts and provided academic guidance to students on difficult topics.

ACADEMIC CREDENTIALS

•Rajshahi University of Engineering & Technology

CGPA: 3.72/4.00

B.Sc. in Electrical and Electronic Engineering

March, 2022 – September, 2026

RESEARCH INTERESTS

Biomedical Signal Processing (ECG, PPG), Non-Invasive Biosensing, Deep Learning & Model Optimization, Applied Machine Learning & Cross-Domain Modeling (Mining Engineering Applications).

PUBLICATIONS

Journal Articles

1. **T Ahmmmod**, MA Motin, and S Kabir, “A Generalizable Framework for Bundle Branch Block Detection Using Knowledge Distillation-Guided Lead-Aware Convolutional Neural Networks and Hybrid Feature Fusion,” *IEEE Transactions on Human-Machine Systems*. (Under Review) [Field: Bio-ML]
2. A Deb*, **T Ahmmmod***, P Roy, PPC Awishi, MA Motin, and A Dash, “Empirically-Driven Machine Learning with Aftershock Decay Optimization: A Generalized Approach for Accurate Ground Vibration Forecasting,” *Engineering with Computers*, Springer Nature. (Under Review) [Field: Cross-ML]

Conference Proceedings

1. **T Ahmmmod**, MT Amin, Z Ferdus, TNH Naem, E Ferdous, and MM Tajwear, “Cross-Group Meta-Ensemble Learning for Heart Disease Classification Using ML, DL, SSL, and Biomedical PLMs Models,” in *Proc. IEEE SPICSCON*, 2026. (Accepted) [Field: Bio-ML]
2. **T Ahmmmod***, R Sarkar*, and MA Motin, “GlucoNova: A Hybrid Model for Non-Invasive Blood Glucose Monitoring Using Near-Infrared Photoplethysmography,” in *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. DOI: 10.1109/ICCIT68739.2025.11490406. [Field: Bio-Device]
3. MT Amin*, **T Ahmmmod***, and TNH Naem, “Integrating Machine Learning Ensembles and Large Language Models for Heart Disease Prediction Using Voting Fusion,” in *Proc. ICECTE*, 2026. DOI: Link. [Field: Bio-ML]
4. A Chaudhury and **T Ahmmmod**, “Point-Supervised Cross-Conditioned Cascade Framework for Explainable Joint Tumor and Modality Classification in Brain MRI,” in *Proc. 29th International Conference on Computer and Information Technology (ICCIT)*, 2026. (Under Review) [Field: Bio-ML]
5. A Deb, **T Ahmmmod**, and S Siddique, “Empirical-Driven Machine Learning and Bayesian Optimization for Uniaxial Compressive Strength Prediction from Bengal Basin,” in *Proc. 29th International Conference on Computer and Information Technology (ICCIT)*, 2026. (Under Review) [Field: Cross-ML]

* indicates equal contribution

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, HTML

Python Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch

Hardware & Embedded Systems: Arduino, Shimmer ECG/PPG/GSR Unit, Multi-Sensors, Proteus, Verilog

Engineering Tools & Software: EasyEDA, Arduino IDE, Kaggle, AutoCAD

Documentation & Design: LaTeX, MS Office (Word, Excel, PowerPoint), Canva

Research Highlights: First Author (2 Journals, 3 Conferences), Attended 3 Conferences

PROJECTS & WORKS

•Multi Signal Database

RUET TechnoFair, 2025

–Description: Developed a deep-learning powered embedded software designed to analyze multi-channel ECG signals, automatically detecting arrhythmias with high precision using the PyTorch framework.

–Tools: Python, PyTorch, Signal Processing, Deep Learning.

•Iontophoresis Device for Hyperhidrosis Treatment

RUET Project 1, 2024

–Description: Designed an automated, safe medical device for non-invasive treatment of hyperhidrosis (excessive sweating).

–Tools: Embedded Systems, Circuit Design, Medical Device, Hyperhidrosis Therapy.

ACHIEVEMENTS & MEMBERSHIPS

• RUET Merit Scholarship

2022 – 2026

Awarded for outstanding academic excellence and maintaining a high CGPA.

• IEEE Student Membership

2022 – Present

Active student member of the Institute of Electrical and Electronics Engineers (IEEE).

• ASSR Membership

Club Member

Active member of Astronomy and Science Society of RUET (ASSR).

REFERENCES

Dr. Mohammad Abdul Motin

PhD, The University of Melbourne, Australia

Associate Professor (Thesis Supervisor)

Department of Electrical and Electronic Engineering

Rajshahi University of Engineering & Technology

Contact: +880 1712863393

E-mail: m.a.motin@ieee.org

Dr. Sumaiya Kabir

PhD, RMIT University, Australia

Professor

Department of Electrical and Electronic Engineering

Rajshahi University of Engineering & Technology

Contact: +880 1324953607

E-mail: sumaiyakabir_09@yahoo.com